Example 8.3

We have $X_t = \theta Z_{t-1} + Z_t$

Seek estimates of θ, σ_Z^2 using method of moments.

Know: $r_1 = -0.25$; $c_0 = 1.25$.

For an MA(1) process we know that

$$\rho(1) = \frac{\theta}{1+\theta^2}$$

For $\tilde{\theta}$: Set $\dfrac{\tilde{\theta}}{1+\tilde{\theta}^2} = r_1 = -\dfrac{1}{4}$.

$$\Rightarrow 4\tilde{\theta} = -(1+\tilde{\theta}^2)$$

$$\Rightarrow \tilde{\theta}^2 + 4\tilde{\theta} + 1 = 0$$

$$\Rightarrow (\tilde{\theta}+2)^2 - 3 = 0$$

$$\Rightarrow (\tilde{\theta}+2)^2 = 3$$

$$\tilde{\theta} + 2 = \pm\sqrt{3}$$

$$\tilde{\theta} = -2 \pm \sqrt{3}.$$

Choose $\tilde{\theta} = -2 + \sqrt{3} \approx 0.268$ (to ensure invertibility).

For MA(1) process

$$\gamma(0) = Var(X_t) = \sigma_Z^2(1+\theta^2)$$

Set $\tilde{\sigma}_Z^2(1+\tilde{\theta}^2) = c_0 \Rightarrow \tilde{\sigma}_Z^2 = \dfrac{c_0}{1+\tilde{\theta}^2}$

$$\tilde{\sigma}_z^2 = \frac{1.25}{1 + 0.268^2} \approx 1.166$$